



CHAPTER

5

Engineering Thermodynamics I

Lecture Notes:

Second law of Thermodynamics



Objective of the chapter

- ❖ Introduce the second law of thermodynamics.
- ❖ Identify valid processes as those that satisfy both the first and second laws of thermodynamics.
- ❖ Discuss thermal energy reservoirs, reversible and irreversible processes, heat engines, refrigerators, and heat pumps.
- ❖ Describe the Kelvin–Planck and Clausius statements of the second law of thermodynamics.
- ❖ Discuss the concepts of perpetual-motion machines.

Cont.....

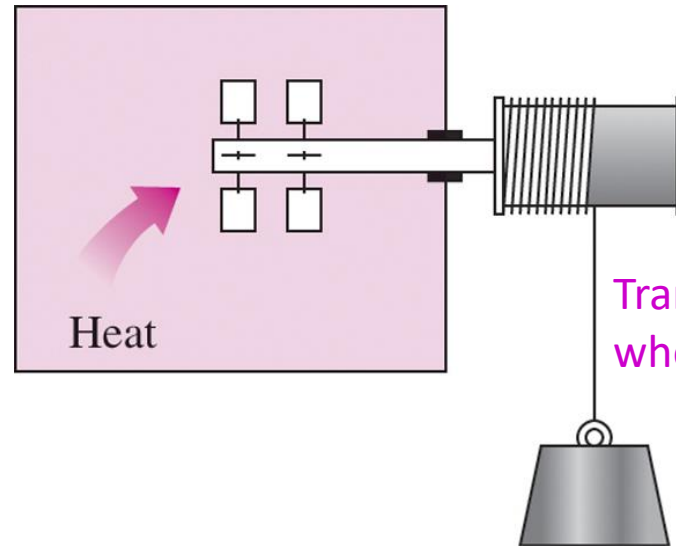
- ❖ Apply the second law of thermodynamics to cycles and cyclic devices.
- ❖ Apply the second law to develop the absolute thermodynamic temperature scale.
- ❖ Describe the Carnot cycle.
- ❖ Examine the Carnot principles, idealized Carnot heat engines, refrigerators, and heat pumps.
- ❖ Determine the expressions for the thermal efficiencies and coefficients of performance for reversible heat engines, heat pumps, and refrigerators

Introduction

- A process must satisfy the first law in order to occur.

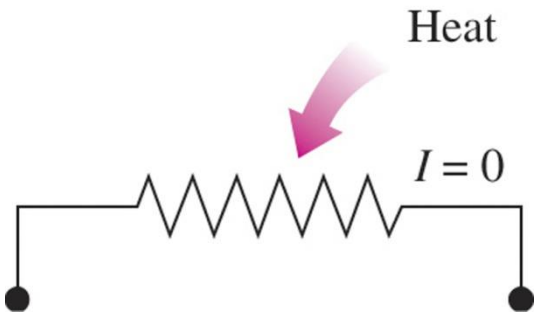


A cup of hot coffee does not get hotter in a cooler room.



Operated by the fall of a mass

Transferring heat to a paddle wheel will not cause it to rotate.



heating of a room by the passage of electric current through a resistor

Transferring heat to a wire will not generate electricity.



Introduction

These processes cannot occur even though they are not in violation of the first law.

- Satisfying the first law alone does not ensure that the process will take place.
- Second law is useful:
 - *provide means for predicting the direction of processes*
 - *determining the best theoretical performance of cycles, heat engines, refrigerators. etc...*

Limitations of first law of thermodynamics

❖ First limitation

- ❖ Limitation:-First law does not help to predict whether the certain process is possible or not.
- ❖ Reason:-It does not specify that heat cannot flow from low temperature body to a high temperature body.

❖ Second limitation

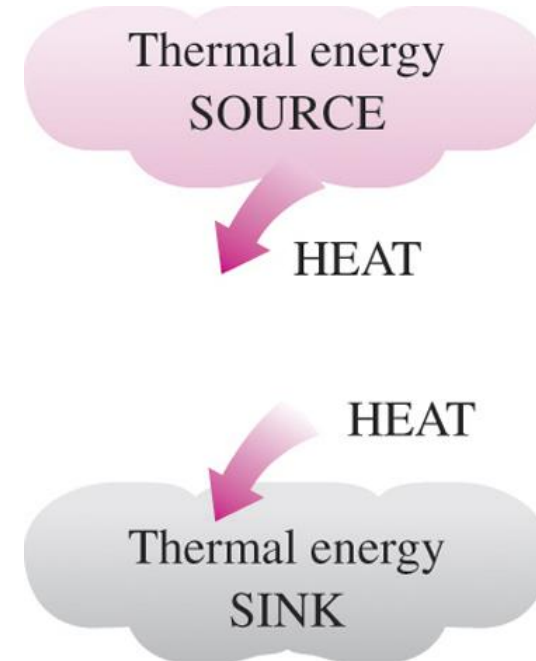
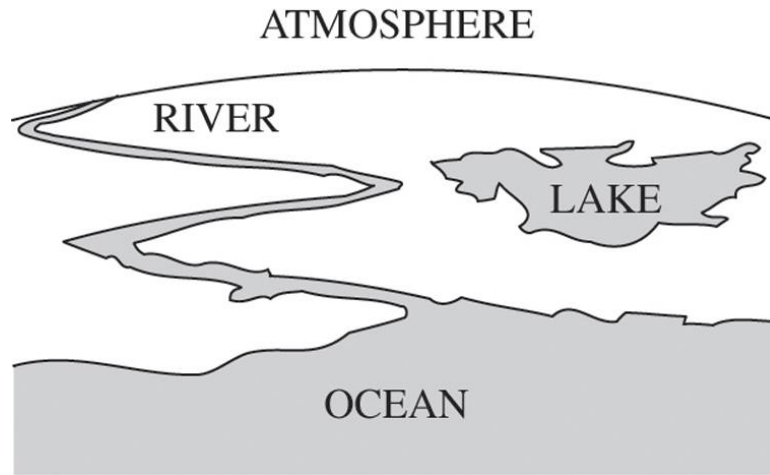
- ❖ Limitation:- The first law does not give information about direction.
- ❖ Reason:- For example, it puts no restriction on the direction of the flow of heat, whether heat can flow from a cold body to a hot body or vice versa.

Limitations of first law of thermodynamics

❖ Third limitation

- ❖ Limitation:- It does not provide and specify sufficient condition for the process to take place.
- ❖ Reason:- This Law is silent about its % of conversion of energy from one form to another form. Work can be converted into equivalent amount of heat but heat cannot be converted into equivalent amount of work.

THERMAL ENERGY RESERVOIRS

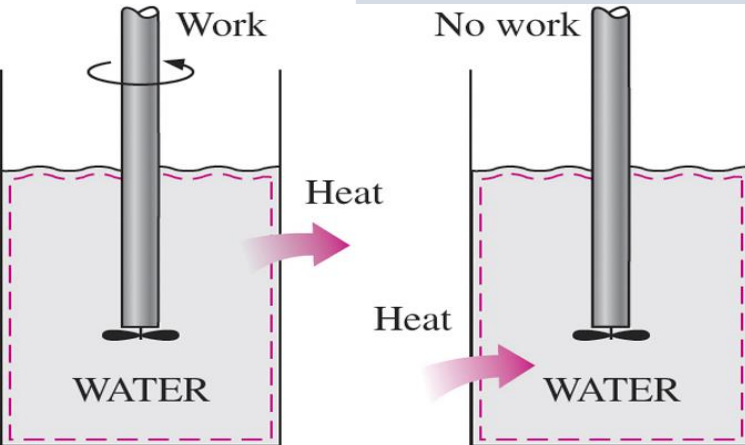


A **source** supplies energy in the form of heat, and a **sink** absorbs it.

Bodies with relatively large thermal masses can be modeled as thermal energy reservoirs.

- A hypothetical body with a relatively **large thermal energy capacity** (mass x specific heat) that can supply or absorb finite amounts of heat without undergoing any change in temperature is called a **thermal energy reservoir**, or just a reservoir.
- In practice, large bodies of water such as **oceans**, **lakes**, and **rivers** as well as the **atmospheric air** can be modeled accurately as thermal energy reservoirs because of their large thermal energy storage capabilities or thermal masses.

Heat Engines



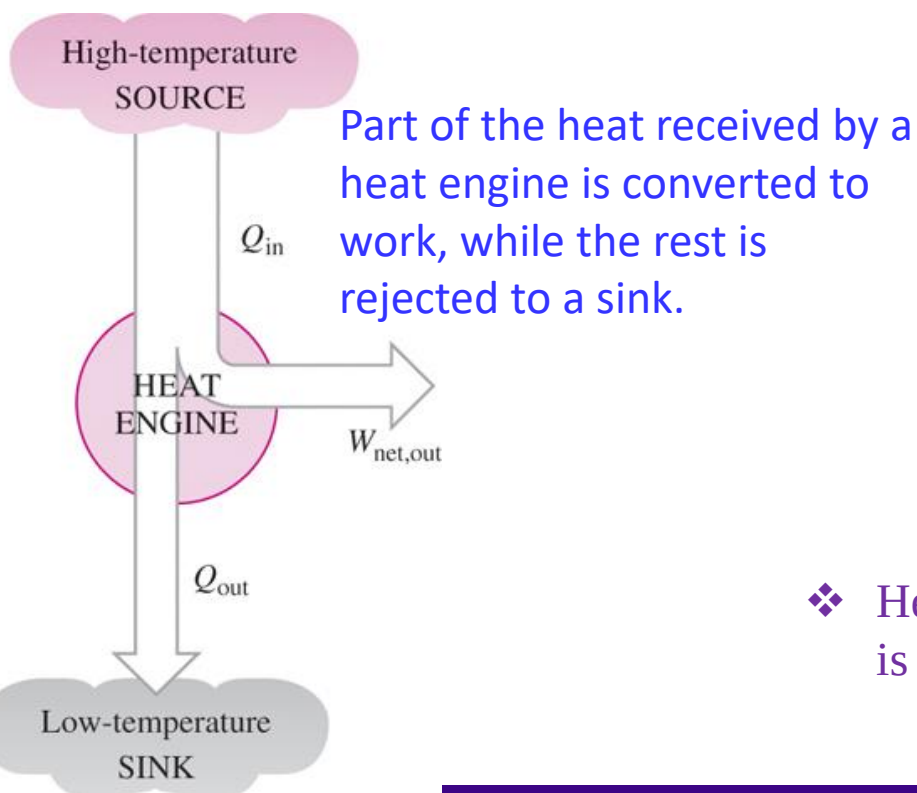
❑ **Work can easily be converted to other forms of energy, but the reverse is not true.**

❖ A heat engine is any closed-cycle device that extracts heat Q_H from a hot reservoir, does useful work, and then exhausts heat Q_C to a cold reservoir.

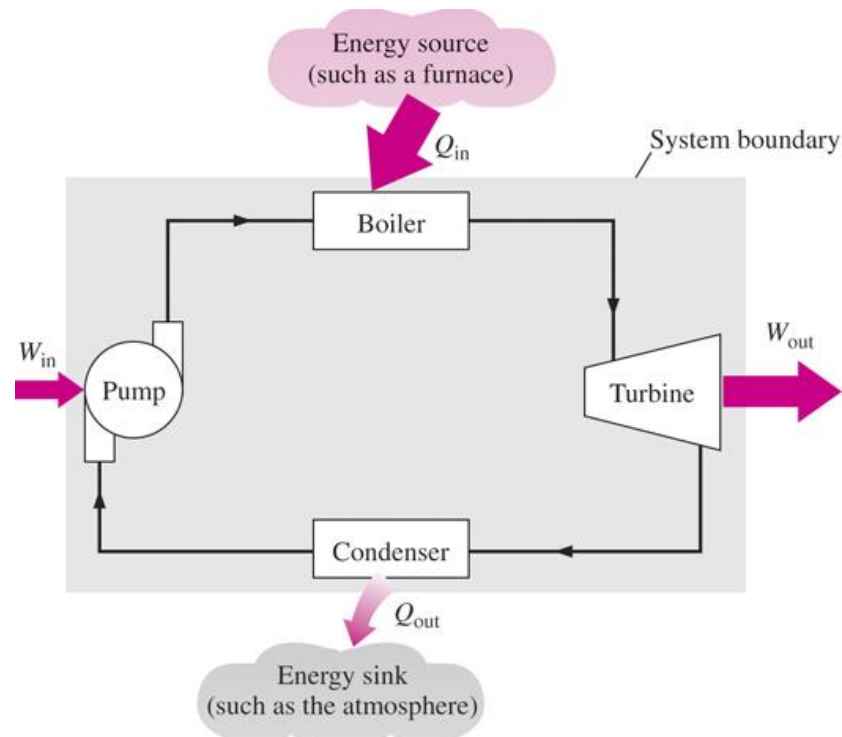
❑ **Heat engine differ considerably from one another, but all can be characterized :**

- they receive heat from a high-temperature source
- they convert part of this heat to work
- they reject the remaining waste heat to a low- temperature sink atmosphere
- they operate on a cycle

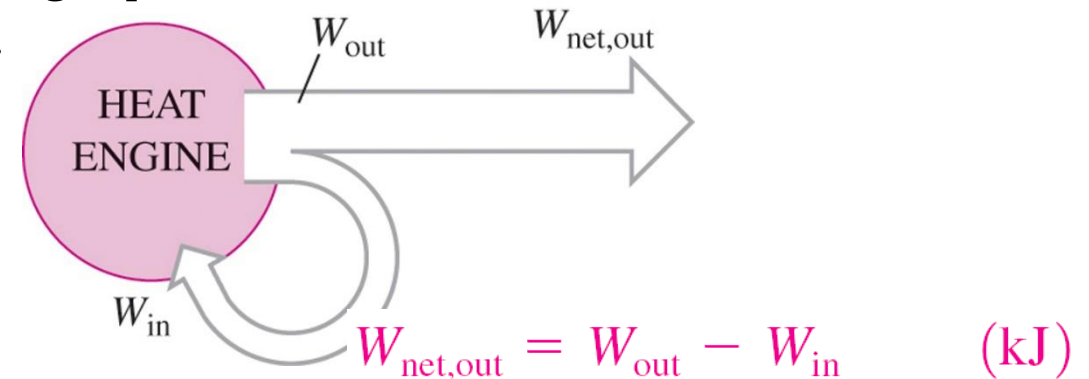
❖ Heat engines and other cyclic devices usually involve a fluid to and from which heat is transferred while undergoing a cycle. This fluid is called the **working fluid**.



A steam power plant



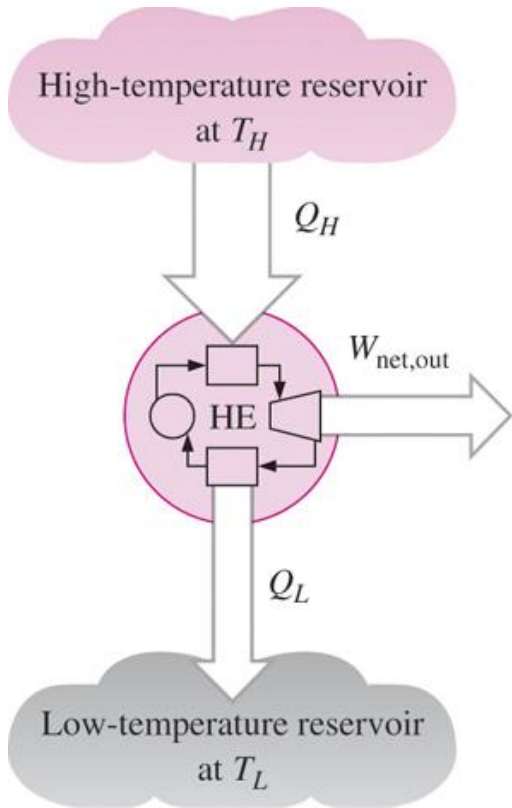
- ❖ Liquid water at ambient temperature is pumped into a boiler at high pressure.
- ❖ Heat from a fuel (heat of combustion of a fossil fuel or heat from a nuclear reaction) is transferred in the boiler to the water, converting it to high-temperature steam at the boiler pressure.
- ❖ Energy is transferred as shaft work from the steam to the surroundings by a device such as a turbine, in which the steam expands to reduced pressure and temperature.
- ❖ Exhaust steam from the turbine is condensed by transfer of heat to the surroundings, producing liquid water for return to the boiler, thus completing the cycle.



A portion of the work output of a heat engine is consumed internally to maintain continuous operation.

$$W_{\text{net,out}} = Q_{\text{in}} - Q_{\text{out}} \quad (\text{kJ})$$

Thermal Efficiency



- ❖ The fraction of the heat input that is converted to the net work output (efficiency = benefit / cost)..
- ❖ Represent the magnitude of the energy wasted in order to complete the cycle.
- ❖ A measure of the performance that is called the thermal efficiency.
- Can be expressed in terms of the desired output and the required input

$$\text{Thermal efficiency} = \frac{\text{Net work output}}{\text{Total heat input}}$$

$$\eta_{th} = \frac{\text{Desired Result}}{\text{Required Input}}$$

- ❖ For a heat engine the desired result is the net work done and the input is the heat supplied to make the cycle operate.



Thermal Efficiency

The thermal efficiency is always less than 1 or less than 100 percent.

where

$$\eta_{th} = \frac{W_{net, out}}{Q_{in}}$$

$$W_{net, out} = W_{out} - W_{in}$$

$$Q_{in} \neq Q_{net}$$

Thermal Efficiency

- ❖ Applying the first law to the cyclic heat engine

$$Q_{net, in} - W_{net, out} = \Delta U$$

$$W_{net, out} = Q_{net, in}$$

$$W_{net, out} = Q_{in} - Q_{out}$$

- ❖ The cycle thermal efficiency may be written as

$$\begin{aligned}\eta_{th} &= \frac{W_{net, out}}{Q_{in}} \\ &= \frac{Q_{in} - Q_{out}}{Q_{in}} \\ &= 1 - \frac{Q_{out}}{Q_{in}}\end{aligned}$$



Thermal Efficiency

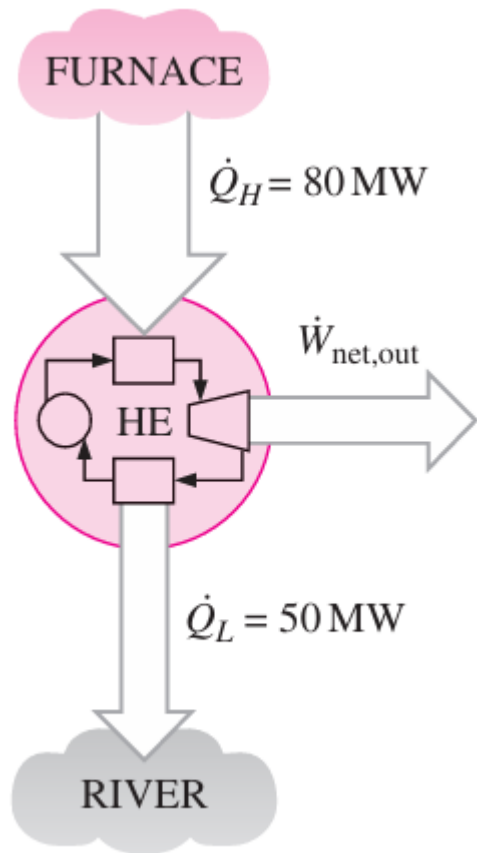
- A *thermodynamic temperature scale* related to the heat transfers between a reversible device and the high and low-temperature reservoirs by

$$\frac{Q_L}{Q_H} = \frac{T_L}{T_H}$$

- The heat engine that operates on the reversible Carnot cycle is called the Carnot Heat Engine in which its efficiency is

$$\eta_{th, rev} = 1 - \frac{T_L}{T_H}$$

Example 5.1



Heat is transferred to a heat engine from a furnace at a rate of 80 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the net power output and the thermal efficiency for this heat engine.

Solution The rates of heat transfer to and from a heat engine are given. The net power output and the thermal efficiency are to be determined.

Assumptions Heat losses through the pipes and other components are negligible.

Analysis A schematic of the heat engine is given in Fig. 6–16. The furnace serves as the high-temperature reservoir for this heat engine and the river as the low-temperature reservoir. The given quantities can be expressed as

$$\dot{Q}_H = 80 \text{ MW} \quad \text{and} \quad \dot{Q}_L = 50 \text{ MW}$$

The net power output of this heat engine is

$$\dot{W}_{\text{net,out}} = \dot{Q}_H - \dot{Q}_L = (80 - 50) \text{ MW} = \mathbf{30 \text{ MW}}$$

Then the thermal efficiency is easily determined to be

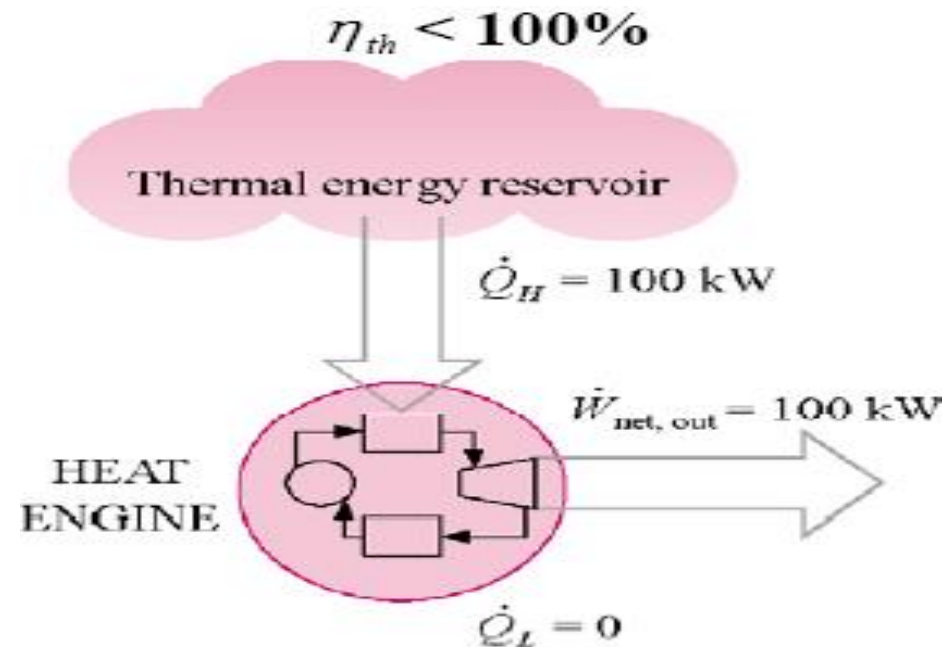
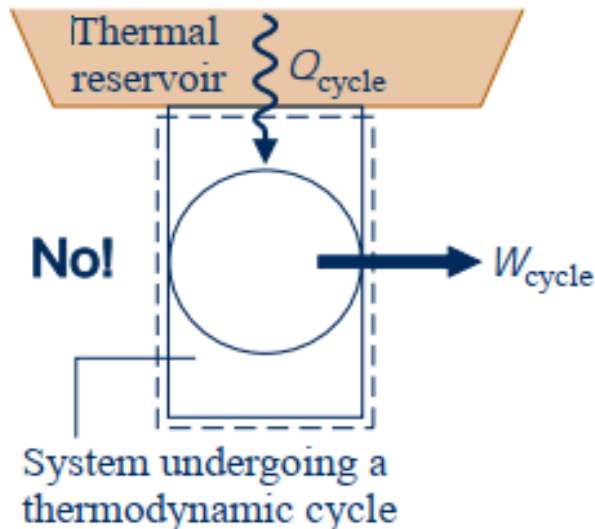
$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net,out}}}{\dot{Q}_H} = \frac{30 \text{ MW}}{80 \text{ MW}} = \mathbf{0.375} \text{ (or 37.5\%)}$$

Discussion Note that the heat engine converts 37.5 percent of the heat it receives to work.

Statements of the Second Law

Kelvin-Planck statement

- ❑ It is impossible for any system to operate in a thermodynamic cycle and deliver a net amount of energy by work to its surroundings while receiving energy by heat transfer from a single thermal reservoir.
- ❑ No heat engine can have a thermal efficiency 100 percent.
- ❑ As for a power plant to operate, the working fluid must exchange heat with the environment as well as the furnace.



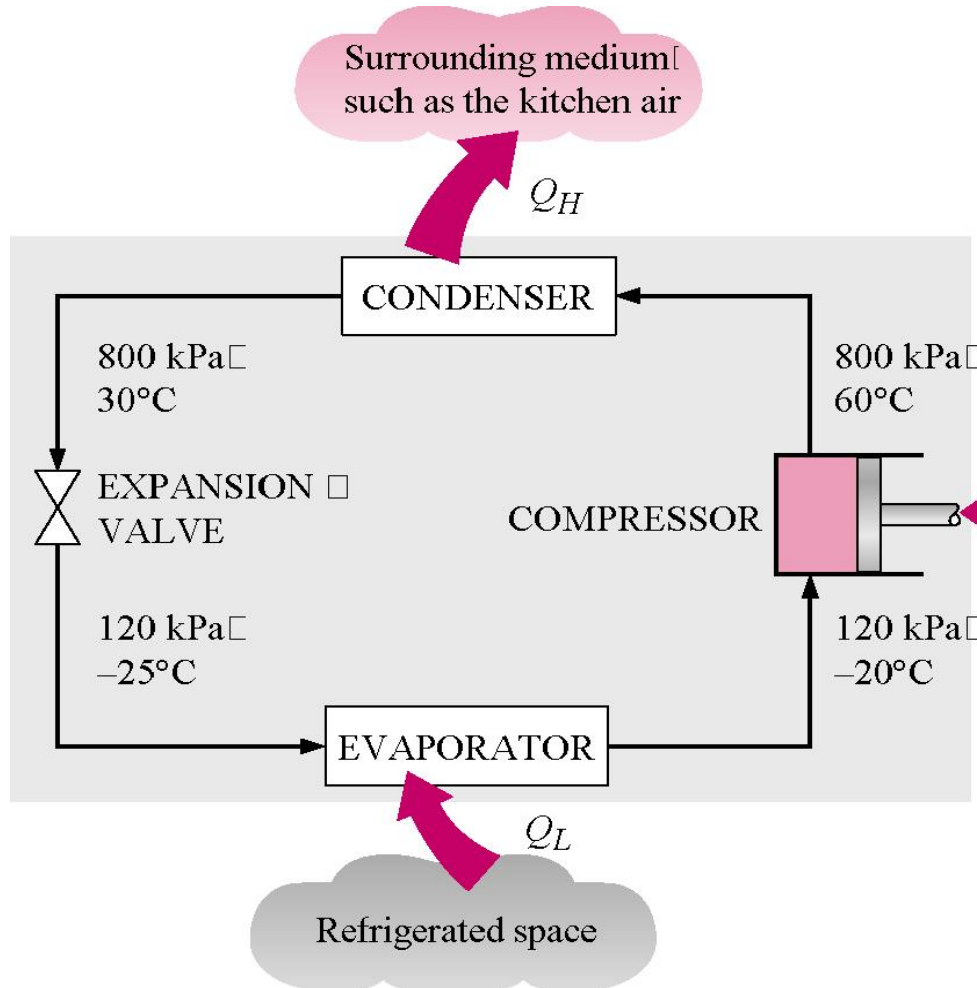


Heat Pumps and Refrigerators

REFRIGERATORS

- ❑ A device that transfers heat from a low temperature medium to a high temperature one is the **refrigerator**.
- ❑ Refrigerators, like heat engines, are cyclic devices.
- ❑ The working fluid used in the refrigeration cycle is called a **refrigerant**.
- ❑ The most frequently used refrigeration cycle is the vapor-compression refrigeration cycle,
- ❑ which involves four main components: a **compressor**, a **condenser**, an **expansion valve**, and an **evaporator**

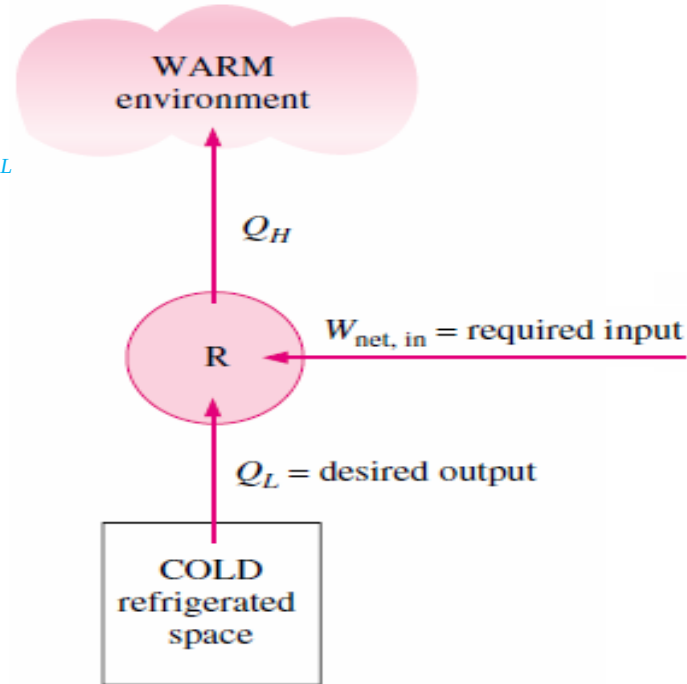
Heat Pumps and Refrigerators



- ❖ The refrigerant enters the compressor as a vapor and is compressed to the condenser pressure.
- ❖ It leaves the compressor at a relatively high temperature and cools down and condenses as it flows through the coils of the condenser by rejecting heat to the surrounding medium.

Heat Pumps and Refrigerators

The objective of a refrigerator is to remove Q_L from the cooled space.



❖ The performance of refrigerators is expressed in terms of the coefficient of performance (COP) which is defined as

$$\text{COP}_R = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_L}{W_{\text{net}, \text{in}}}$$

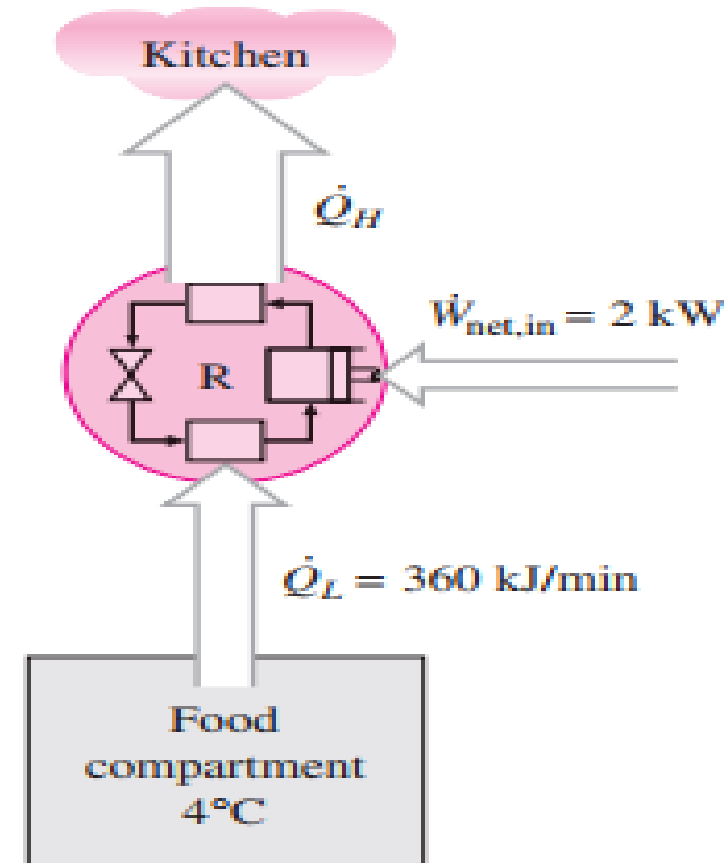
$$W_{\text{net}, \text{in}} = Q_H - Q_L \quad (\text{kJ})$$

$$\text{COP}_R = \frac{Q_L}{Q_H - Q_L} = \frac{1}{Q_H/Q_L - 1}$$

Heat Pumps and Refrigerators

Example 5.2

The food compartment of a refrigerator, shown in Fig below, is maintained at 4°C by removing heat from it at a rate of 360 kJ/min . If the required power input to the refrigerator is 2 kW , determine (a) the coefficient of performance of the refrigerator and (b) the rate of heat rejection to the room that houses the refrigerator.



Solution The power consumption of a refrigerator is given. The COP and the rate of heat rejection are to be determined.

Assumptions Steady operating conditions exist.

Analysis (a) The coefficient of performance of the refrigerator is

$$\text{COP}_R = \frac{\dot{Q}_L}{\dot{W}_{\text{net,in}}} = \frac{360 \text{ kJ/min}}{2 \text{ kW}} \left(\frac{1 \text{ kW}}{60 \text{ kJ/min}} \right) = 3$$

That is, 3 kJ of heat is removed from the refrigerated space for each kJ of work supplied.

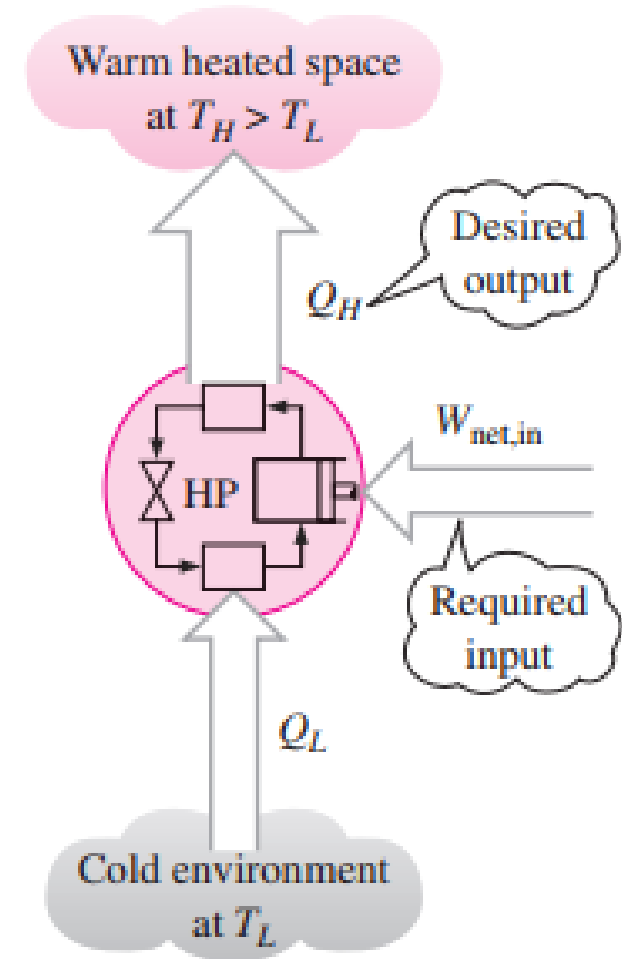
(b) The rate at which heat is rejected to the room that houses the refrigerator is determined from the conservation of energy relation for cyclic devices,

$$\dot{Q}_H = \dot{Q}_L + \dot{W}_{\text{net,in}} = 360 \text{ kJ/min} + (2 \text{ kW}) \left(\frac{60 \text{ kJ/min}}{1 \text{ kW}} \right) = 480 \text{ kJ/min}$$

Heat Pumps and Refrigerators

HEAT PUMPS

- ❑ Another device that transfers heat from a low-temperature medium to a high-temperature one is the **heat pump**.
- ❑ Refrigerators and heat pumps operate on the same cycle but differ in their objectives.

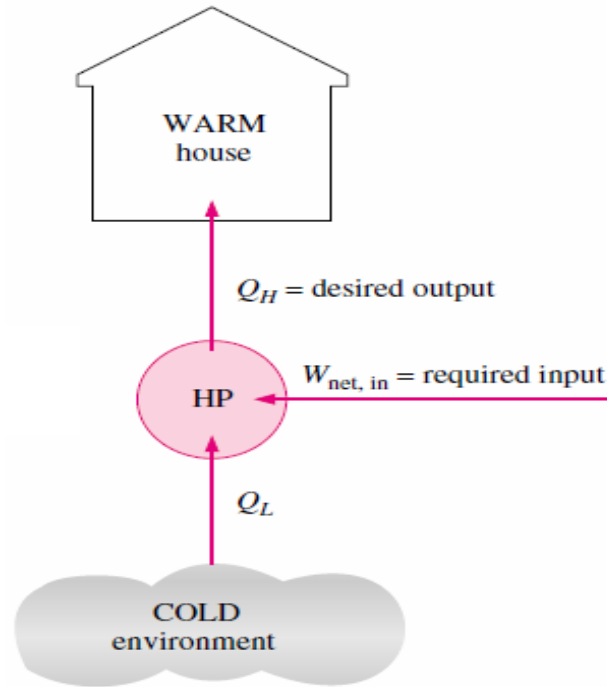




Heat Pumps and Refrigerators

- ❑ The objective of a refrigerator is to maintain the refrigerated space at a low temperature by removing heat from it.
- ❑ Discharging this heat to a higher-temperature medium is merely a necessary part of the operation, not the purpose.
- ❑ The objective of a heat pump, however, is to maintain a heated space at a high temperature.
- ❑ This is accomplished by absorbing heat from a low-temperature source, such as well water or cold outside air in winter, and supplying this heat to the high-temperature medium such as a house

Heat Pumps and Refrigerators



The objective of a heat pump is to supply heat Q_H into the warmer space.

- ❖ The performance of heat pumps is expressed in terms of the coefficient of performance (COP) which is defined as

$$\text{COP}_{\text{HP}} = \frac{\text{Desired output}}{\text{Required input}} = \frac{Q_H}{W_{\text{net, in}}}$$

$$W_{\text{net, in}} = Q_H - Q_L \quad (\text{kJ})$$

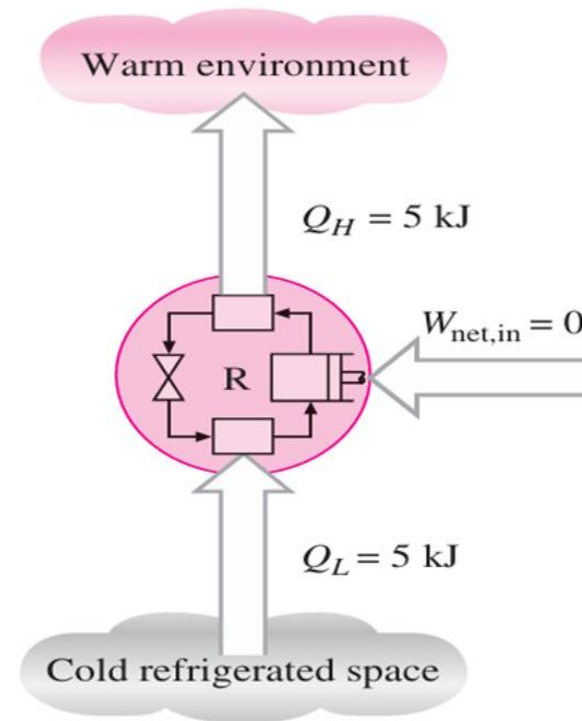
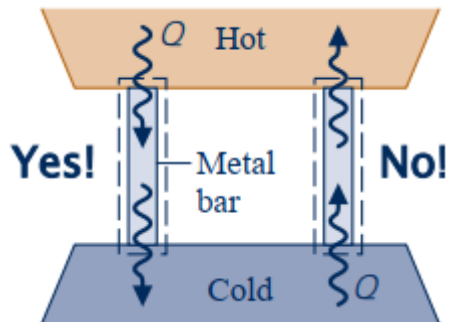
$$\text{COP}_{\text{HP}} = \frac{Q_H}{Q_H - Q_L} = \frac{1}{1 - Q_L/Q_H}$$

$$\text{COP}_{\text{HP}} = \text{COP}_{\text{R}} + 1$$

Statements of the Second Law

Clausius Statement

- ❑ It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to higher-temperature body.
- ❑ In other words, a refrigerator will not operate unless its compressor is driven by an external power source



A refrigerator that violates the Clausius statement of the second law.



Heat Pumps and Refrigerators

Example 5.3

A heat pump is used to meet the heating requirements of a house and maintain it at 20°C . On a day when the outdoor air temperature drops to -2°C , the house is estimated to lose heat at a rate of $80,000 \text{ kJ/h}$. If the heat pump under these conditions has a COP of 2.5, determine (a) the power consumed by the heat pump and (b) the rate at which heat is absorbed from the cold outdoor air.

Solution The COP of a heat pump is given. The power consumption and the rate of heat absorption are to be determined.

Assumptions Steady operating conditions exist.

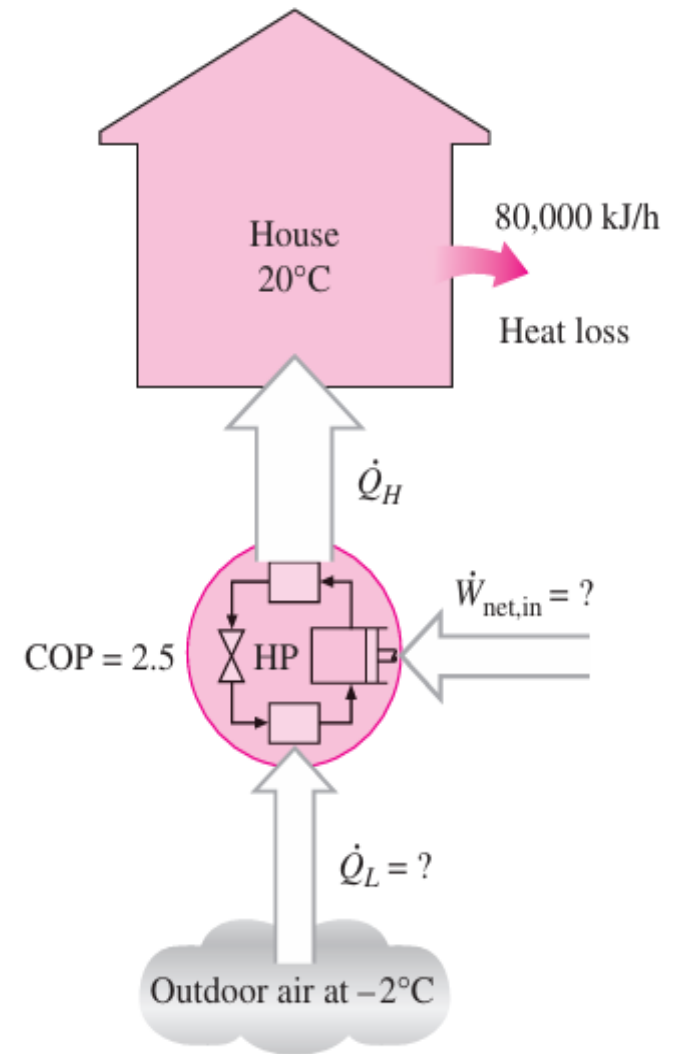
Analysis (a) The power consumed by this heat pump, shown in Fig. is determined from the definition of the coefficient of performance to be

$$\dot{W}_{\text{net,in}} = \frac{\dot{Q}_H}{\text{COP}_{\text{HP}}} = \frac{80,000 \text{ kJ/h}}{2.5} = \mathbf{32,000 \text{ kJ/h}} \text{ (or 8.9 kW)}$$

(b) The house is losing heat at a rate of 80,000 kJ/h. If the house is to be maintained at a constant temperature of 20°C, the heat pump must deliver

heat to the house at the same rate, that is, at a rate of 80,000 kJ/h. Then the rate of heat transfer from the outdoor becomes

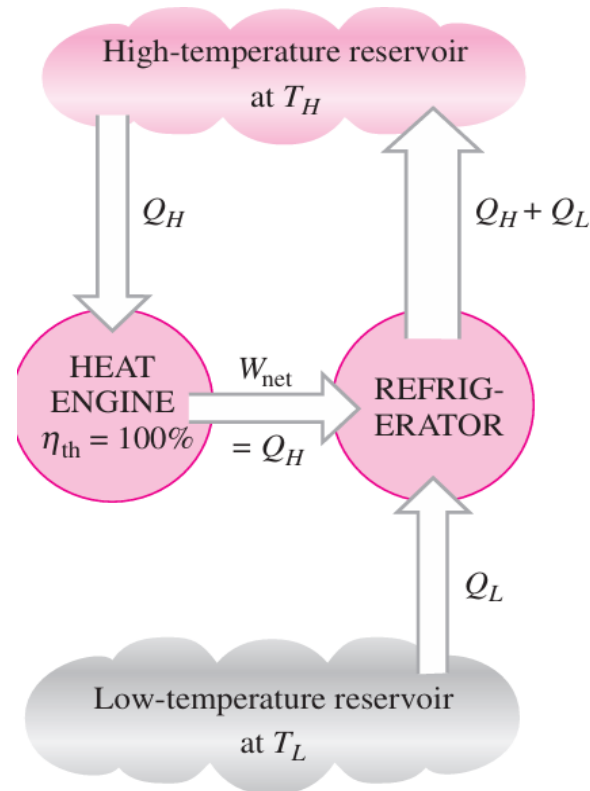
$$\dot{Q}_L = \dot{Q}_H - \dot{W}_{\text{net,in}} = (80,000 - 32,000) \text{ kJ/h} = \mathbf{48,000 \text{ kJ/h}}$$



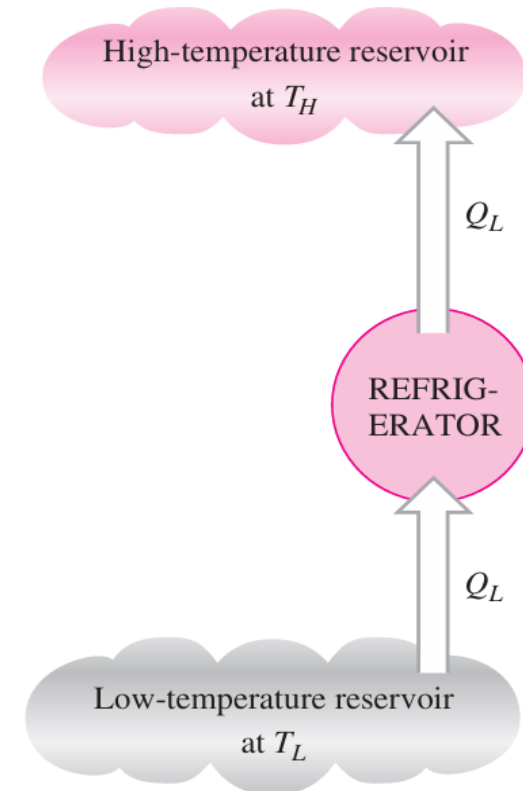
Equivalence of the Two Statements

- The **Kelvin–Planck** and the **Clausius statements** are equivalent in their consequences, and either statement can be used as the expression of the second law of thermodynamics.
- Any device that violates the **Kelvin–Planck** statement also violates the **Clausius statement**, and vice versa.

Proof that the violation of the Kelvin–Planck statement leads to the violation of the Clausius statement.



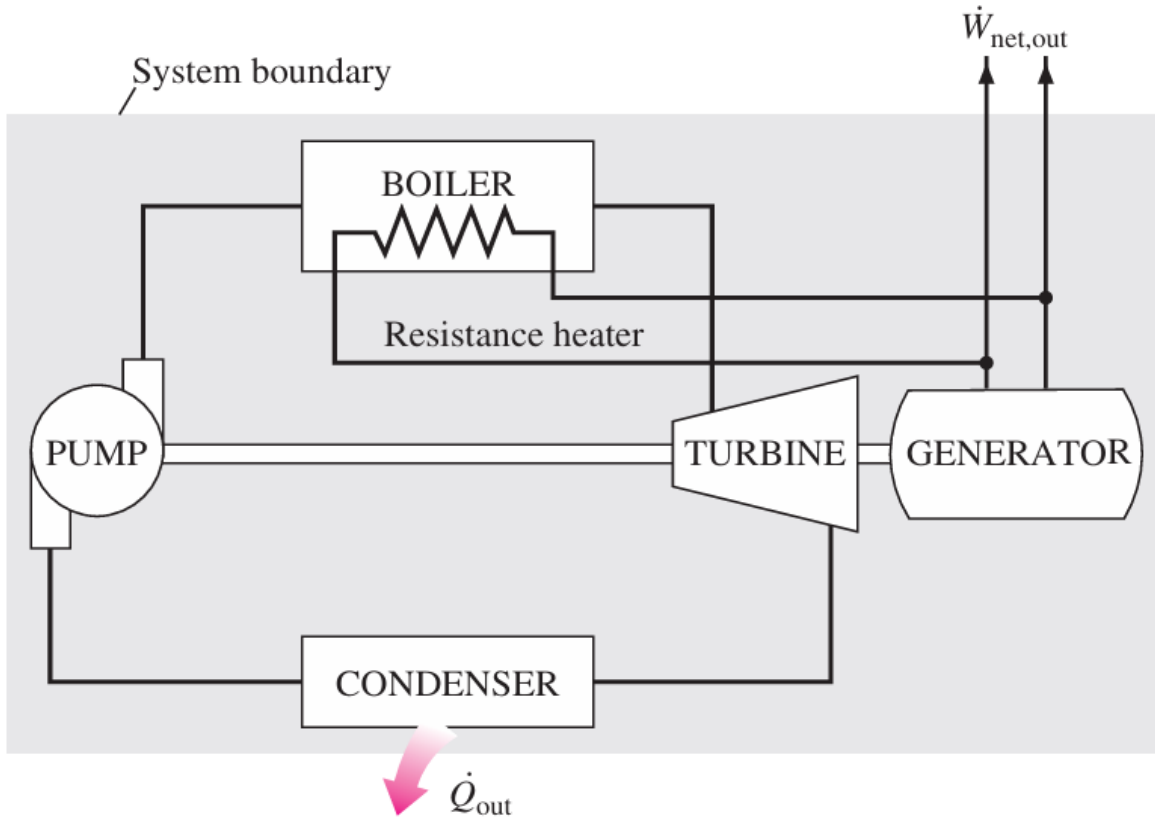
(a) A refrigerator that is powered by a 100 percent efficient heat engine



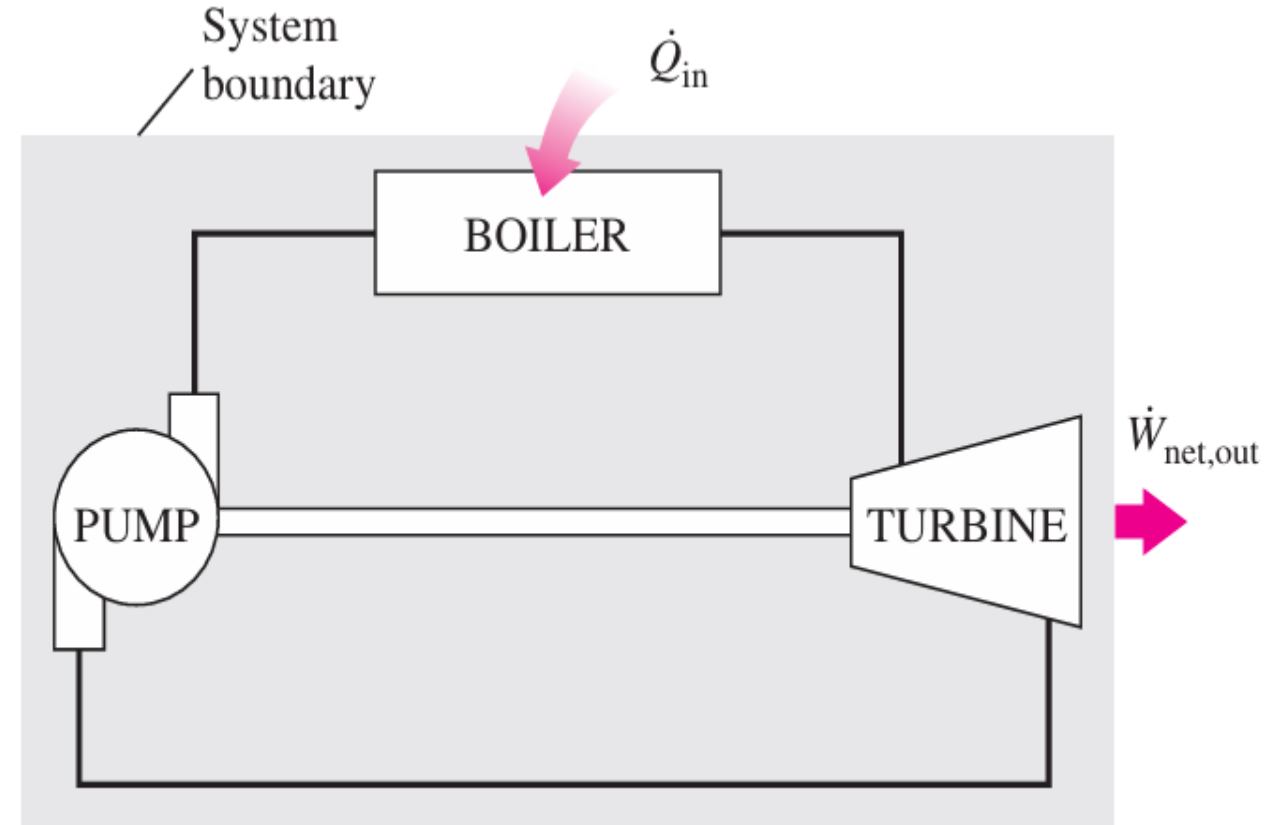
(b) The equivalent refrigerator

PERPETUAL-MOTION MACHINES

- Any device that violates either law is called **a perpetual-motion machine**, and despite numerous attempts, no perpetual-motion machine is known to have worked. But this has not stopped inventors from trying to create new ones.
- A device that violates the first law of thermodynamics (by creating energy) is called **a perpetual-motion machine of the first kind (PMM1)**
- A device that violates the second law of thermodynamics is called a **perpetual-motion machine of the second kind (PMM2)**.



A perpetual-motion machine that violates the first law of thermodynamics (**PMM1**).



A perpetual-motion machine that violates the second law of thermodynamics (**PMM2**).



Reversible and Irreversible Processes

- ❑ The second law of thermodynamics enables us to divide all processes into two classes:
 - ✓ Reversible or ideal process.
 - ✓ Irreversible or natural process.
- ❑ A reversible process is defined as a process that can be reversed without leaving any trace on the surroundings.
- ❑ It means both system and surroundings are returned to their initial states at the end of the reverse process.
- ❑ Processes that are not reversible are called irreversible.
- ❑ Reversible processes do not occur and they are only idealizations of actual processes

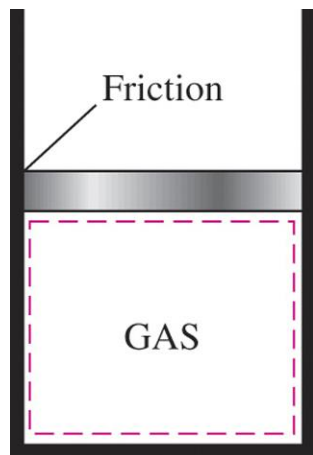


Reversible and Irreversible Processes

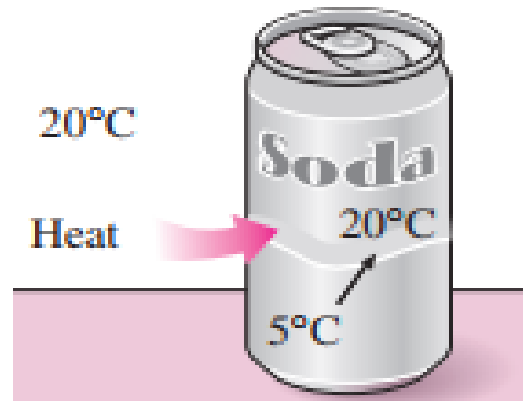
- ❑ We use reversible process concept because,
 - a) they are easy to analyze (since system passes through a series of equilibrium states);
 - b) they serve as limits (idealized models) to which the actual processes can be compared.
- ❑ A process is called internally reversible if no irreversibilities occur within the boundaries of the system during the process.
- ❑ A process is called externally reversible if no irreversibilities occur outside the system boundaries during the process.

Reversible and Irreversible Processes

- ❑ The factors that cause a process to be irreversible are called irreversibility.
- ❑ They include
 - friction, unrestrained expansion,
 - mixing of two fluids, heat transfer across a finite temperature difference,
 - electric resistance, inelastic deformation of solids, and chemical reactions.
- ❑ The presence of any of these effects renders a process irreversible.



Friction renders a process irreversible.



An irreversible heat transfer process



Reversible and Irreversible Processes

□ For many analyses it is convenient to divide the irreversibilities present into two classes,

- ❖ Internal irreversibilities are those that occur within the system.
- ❖ External irreversibilities are those that occur within the surroundings, often the immediate surroundings.



Carnot Cycle

- ❑ If a thermal efficiency of 100% is not possible for heat engines, what then determines the upper limit?
- ❑ One would certainly expect the thermal efficiency of a heat engine to depend on the degree of reversibility of its operation.
- ❑ Indeed, a heat engine operating in a completely reversible manner is very special, and is called a Carnot engine.

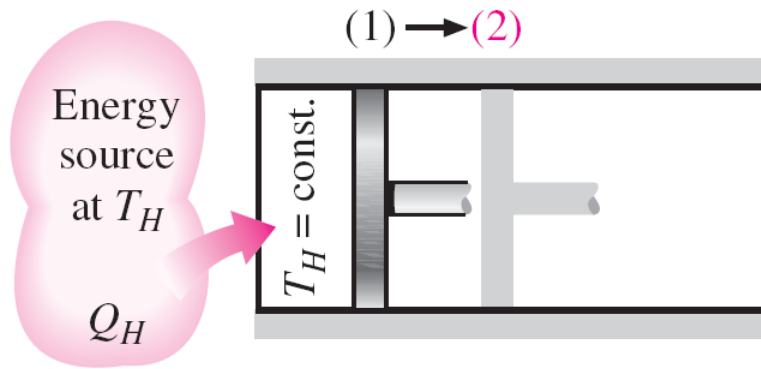


Carnot Cycle

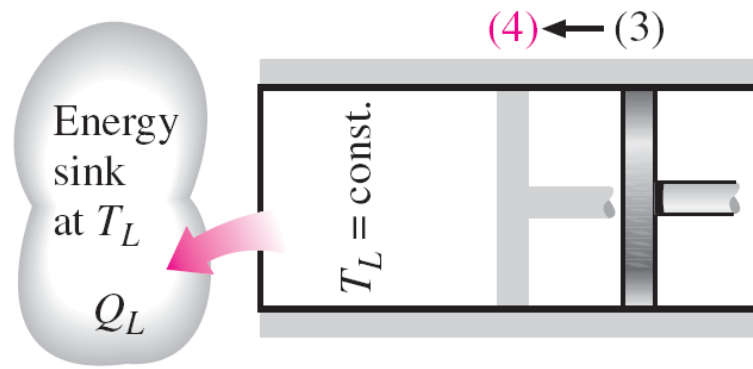
- ❑ The efficiency of a heat-engine cycle greatly depends on how the individual processes that make up the cycle are executed. The net work (or efficiency) can be maximized by using reversible processes.
- ❑ The best known reversible cycle is the Carnot cycle.
- ❑ The theoretical heat engine that operates on the Carnot cycle is called the Carnot heat engine.
- ❑ The Carnot cycle is composed of four reversible processes two isothermal and two adiabatic and it can be executed either in a closed or a steady-flow system.

Carnot Cycle

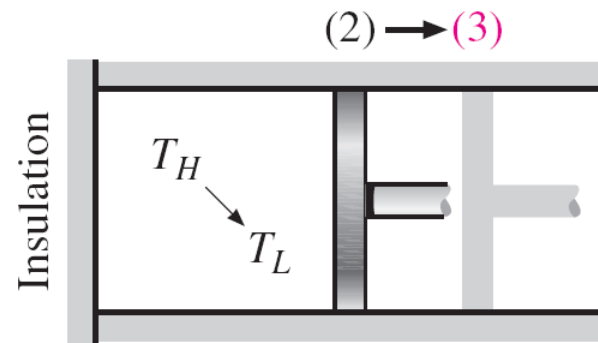
Process	Description
1-2	Reversible isothermal heat addition at high temperature
2-3	Reversible adiabatic expansion from high temperature to low temperature
3-4	Reversible isothermal heat rejection at low temperature
4-1	Reversible adiabatic compression from low temperature to high temperature



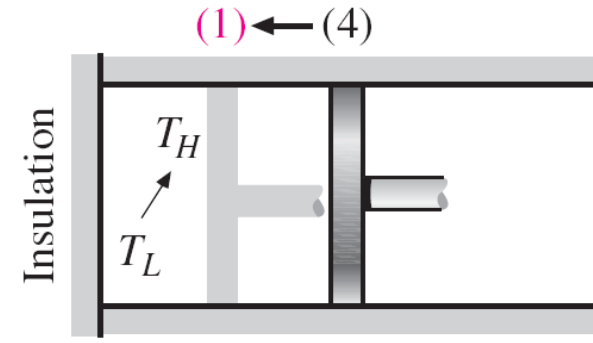
(a) Process 1-2



(c) Process 3-4

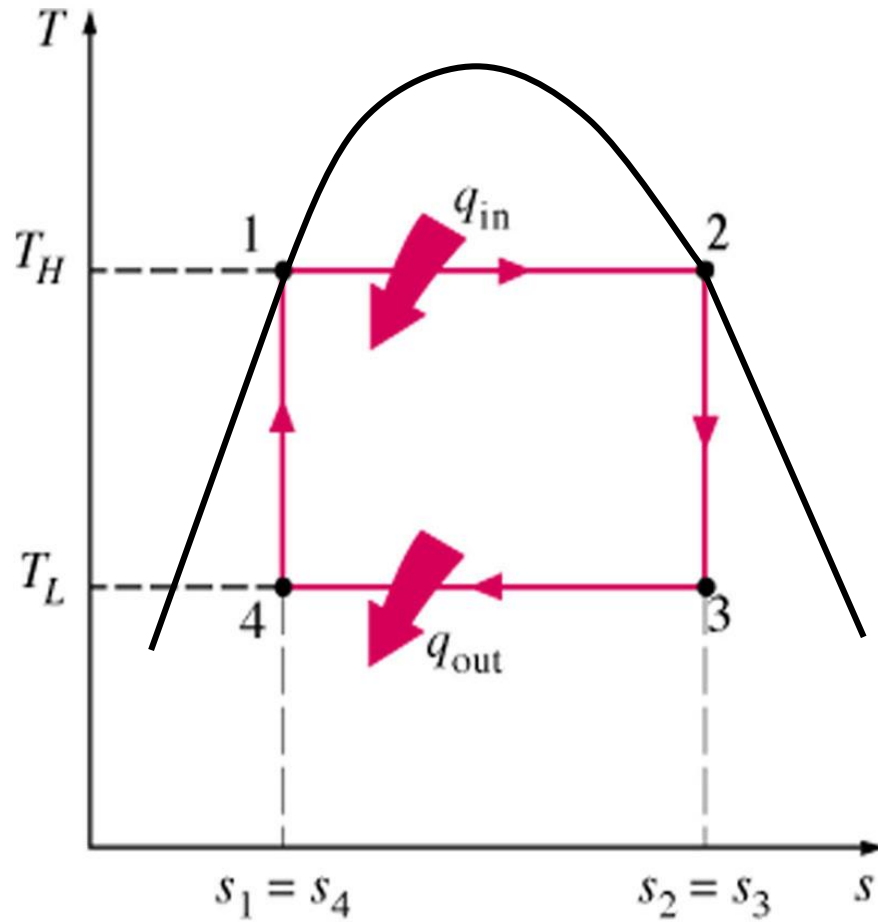


(b) Process 2-3

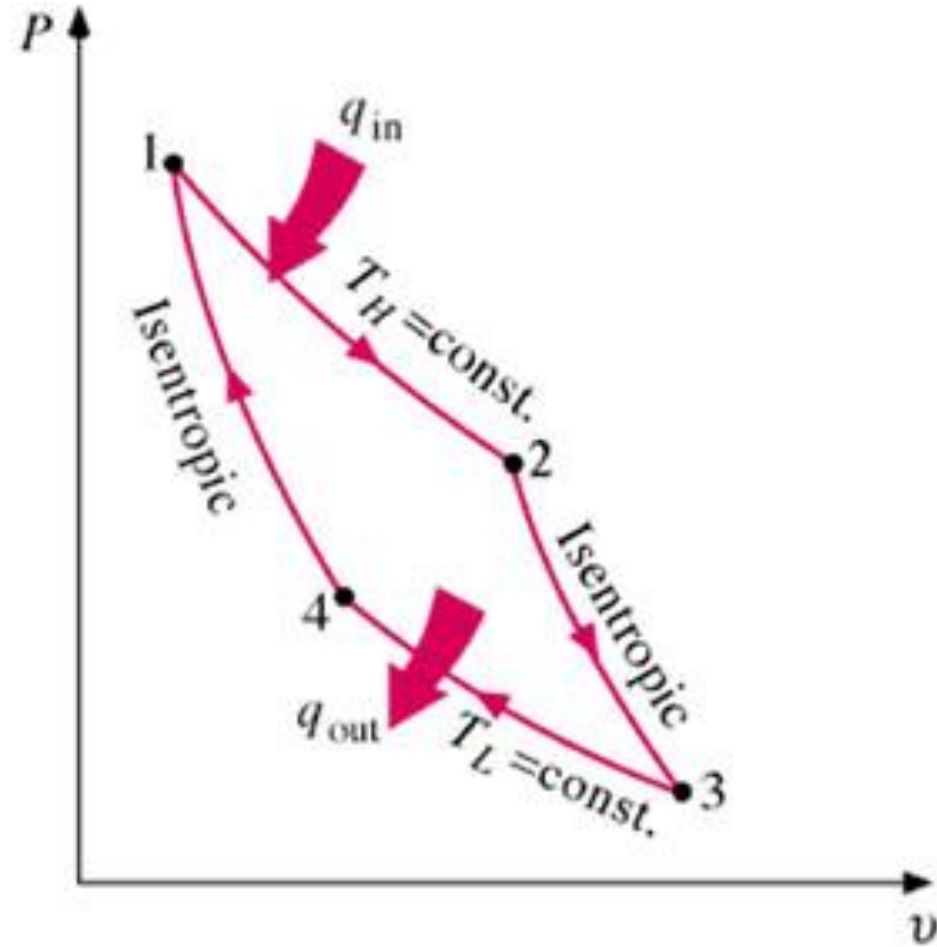


(d) Process 4-1

Execution of Carnot cycle in a piston cylinder device



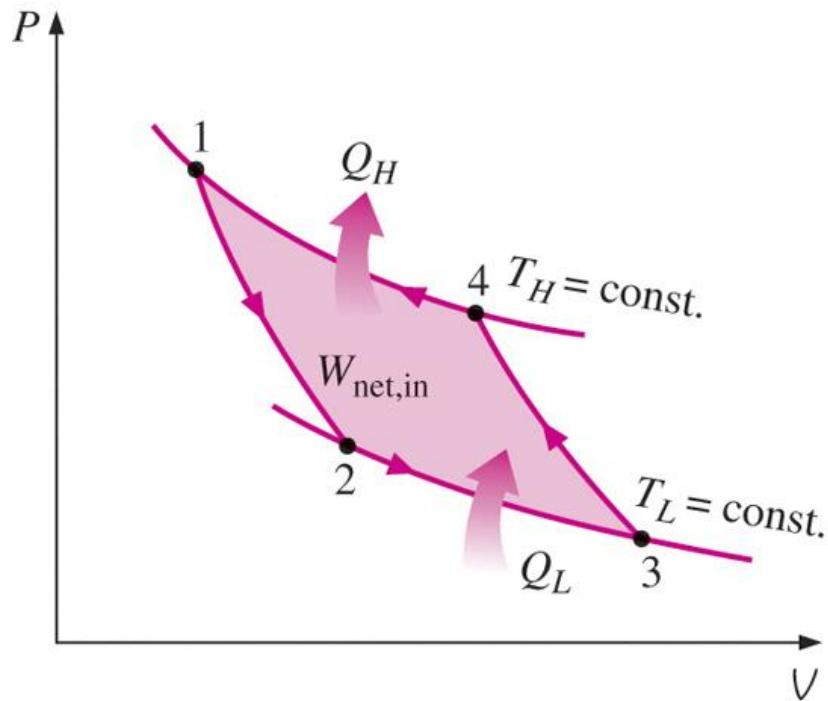
T - s diagram of the Carnot cycle.



P - V diagram of the Carnot cycle.

The Reversed Carnot Cycle

- ❑ The Carnot heat-engine cycle is a totally reversible cycle.
- ❑ Therefore, all the processes that comprise it can be *reversed*, in which case it becomes the **Carnot refrigeration cycle**

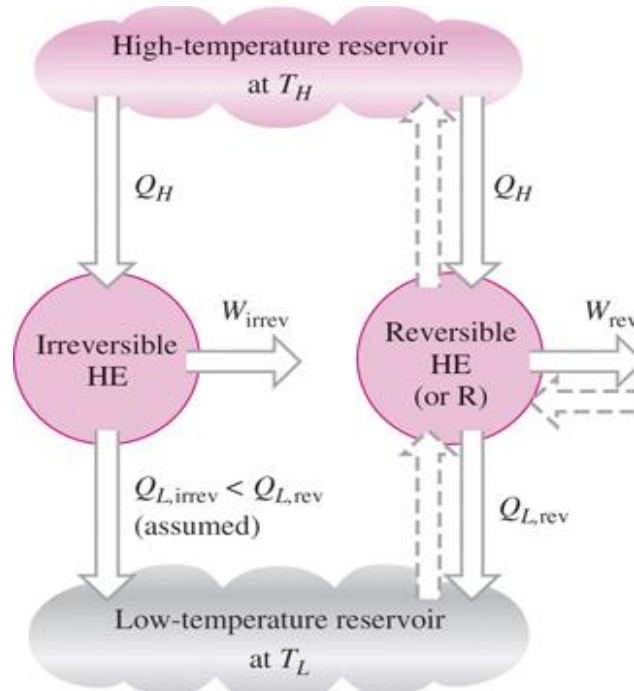
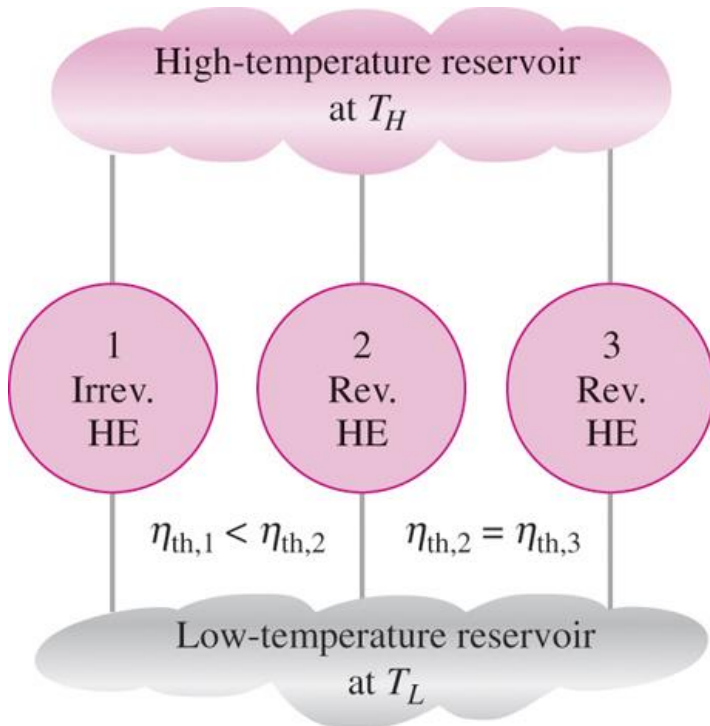


The P-V diagram of the reversed Carnot cycle is the same as the one given for the Carnot cycle, except that the directions of the processes are reversed

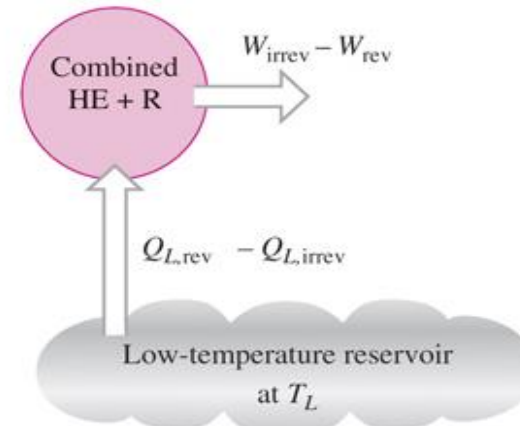
P-V diagram of the reversed Carnot cycle.

The Carnot Principles

- ❑ The efficiency of an irreversible heat engine is always less than the efficiency of a reversible one operating between the same two reservoirs.
- ❑ The efficiencies of all reversible heat engines operating between the same two reservoirs are the same.



(a) A reversible and an irreversible heat engine operating between the same two reservoirs (the reversible heat engine is then reversed to run as a refrigerator)



(b) The equivalent combined system

Proof of the first Carnot principle.



- ❑ The thermal efficiencies of actual and reversible heat engines operating between the same temperature limits compare as follows

$$\eta_{th} \begin{cases} < \eta_{th, rev} & \text{irreversible heat engine} \\ = \eta_{th, rev} & \text{reversible heat engine} \\ > \eta_{th, rev} & \text{impossible heat engine} \end{cases}$$

- ❑ The coefficients of performance of actual and reversible refrigerators operating between the same temperature limits compare as follows

$$COP_R \begin{cases} < COP_{R, rev} & \text{irreversible refrigerator} \\ = COP_{R, rev} & \text{reversible refrigerator} \\ > COP_{R, rev} & \text{impossible refrigerator} \end{cases}$$

Example 5.4

A steam power plant produces 50 MW of net work while burning fuel to produce 150 MW of heat energy at the high temperature. Determine the cycle thermal efficiency and the heat rejected by the cycle to the surroundings.

Solution:

$$\begin{aligned}\eta_{th} &= \frac{W_{net, out}}{Q_H} \\ &= \frac{50 \text{ MW}}{150 \text{ MW}} = 0.333 \quad \text{or} \quad 33.3\%\end{aligned}$$

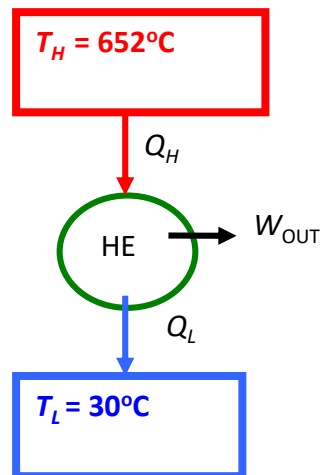
$$\begin{aligned}W_{net, out} &= Q_H - Q_L \\ Q_L &= Q_H - W_{net, out} \\ &= 150 \text{ MW} - 50 \text{ MW} \\ &= 100 \text{ MW}\end{aligned}$$

Example 5.5

A Carnot heat engine receives 500 kJ of heat per cycle from a high-temperature heat reservoir at 652°C and rejects heat to a low-temperature heat reservoir at 30°C. Determine :

- (a) The thermal efficiency of this Carnot engine
- (b) The amount of heat rejected to the low-temperature heat reservoir

Solution:



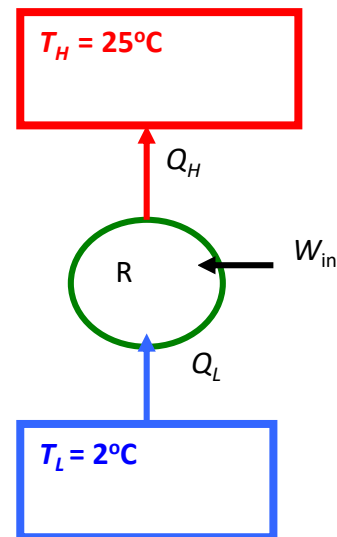
$$\begin{aligned}
 \eta_{th, rev} &= 1 - \frac{T_L}{T_H} \\
 &= 1 - \frac{(30 + 273)K}{(652 + 273)K} \\
 &= 0.672 \quad \text{or} \quad 67.2\%
 \end{aligned}$$

$$\begin{aligned}
 \frac{Q_L}{Q_H} &= \frac{T_L}{T_H} \\
 &= \frac{(30 + 273)K}{(652 + 273)K} = 0.328 \\
 Q_L &= 500 \text{ kJ} (0.328) \\
 &= 164 \text{ kJ}
 \end{aligned}$$

Example 5.6

An inventor claims to have developed a refrigerator that maintains the refrigerated space at 2°C while operating in a room where the temperature is 25°C and has a COP of 13.5. Is there any truth to his claim?

Solution:



$$\begin{aligned} COP_R &= \frac{Q_L}{Q_H - Q_L} = \frac{T_L}{T_H - T_L} \\ &= \frac{(2 + 273)K}{(25 - 2)K} \\ &= 11.96 \end{aligned}$$

- this claim is also false!



Supplementary Problem 5.1

1. A 600 MW steam power plant, which is cooled by a river, has a thermal efficiency of 40 percent. Determine the rate of heat transfer to the river water. Will the actual heat transfer rate be higher or lower than this value? Why?

[900 MW]

2. A steam power plant receives heat from a furnace at a rate of 280 GJ/h. Heat losses to the surrounding air from the steam as it passes through the pipes and other components are estimated to be about 8 GJ/h. If the waste heat is transferred to the cooling water at a rate of 145 GJ/h, determine (a) net power output and (b) the thermal efficiency of this power plant.

[35.3 MW, 45.4%]

3. An air conditioner removes heat steadily from a house at a rate of 750 kJ/min while drawing electric power at a rate of 6 kW. Determine (a) the COP of this air conditioner and (b) the rate of heat transfer to the outside air.

[2.08, 1110 kJ/min]



4. Determine the COP of a heat pump that supplies energy to a house at a rate of 8000 kJ/h for each kW of electric power it draws. Also, determine the rate of energy absorption from the outdoor air.
[2.22, 4400 kJ/h]
5. An inventor claims to have developed a heat engine that receives 700 kJ of heat from a source at 500 K and produces 300 kJ of net work while rejecting the waste heat to a sink at 290 K. Is this reasonable claim?
6. An air-conditioning system operating on the reversed Carnot cycle is required to transfer heat from a house at a rate of 750 kJ/min to maintain its temperature at 24°C. If the outdoor air temperature is 35°C, determine the power required to operate this air-conditioning system.
[0.463 kW]
7. A heat pump is used to heat a house and maintain it at 24°C. On a winter day when the outdoor air temperature is -5°C, the house is estimated to lose heat at a rate of 80,000 kJ/h. Determine the minimum power required to operate this heat pump.

[2.18 kW]



THANK YOU